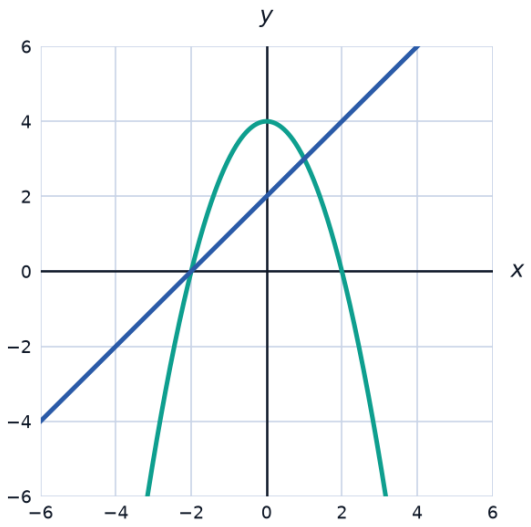
Vectors and Embeddings in AI

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

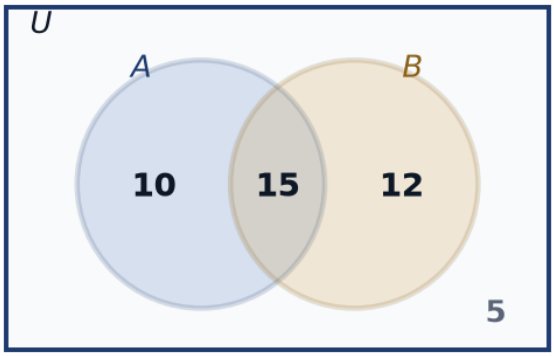
1 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Two document embeddings $\mathbf{a} = [5, 10]^T$ and $\mathbf{b} = [10, 5]^T$ are compared. If $||\mathbf{a}|| = ||\mathbf{b}|| = \sqrt{125}$, calculate their exact cosine similarity.



2 [Foundational] • Target: Vector Spaces and Linear Combinations

Consider a dataset of n data points, where each data point is a vector in $\mathbb{R}^D$. A non-linear dimensionality reduction technique maps these data points to a lower-dimensional space $\mathbb{R}^d$. The mapping function for a single data point $\mathbf{x} \in \mathbb{R}^D$ is given by $f(\mathbf{x}) = \phi(\mathbf{x})$, where $\phi : \mathbb{R}^D \to \mathbb{R}^d$ is a function designed to preserve local neighborhood structures. Suppose the mapping ϕ is defined by an explicit feature map $\phi(\mathbf{x}) = [\mathbf{x}^T\mathbf{x}, \mathbf{x}^T\mathbf{x}, \ldots, \mathbf{x}^T\mathbf{x}]^T$ where the vector $\mathbf{x} \in \mathbb{R}^{10}$ and d is the dimension of the embedded space. If the goal is to embed these points into a space such that pairwise Euclidean distances are preserved as closely as possible under a specific distortion, and it is found that $d = 15$ yields optimal results for this mapping under a certain constraint, determine the effective rank of the feature map's image space.



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3 [Foundational] • Target: Vector Norms and Sparsity

Consider a linear transformation represented by the matrix $M = [2, k; 1, 3]$. A vector $\mathbf{v} = [1; 2]$ is transformed by M into $\mathbf{v}' = M\mathbf{v} = [8; 7]$. Determine the value of the unknown parameter k.

4 [Foundational] • Target: Dot Products, Cosine Similarity and Angular Distance

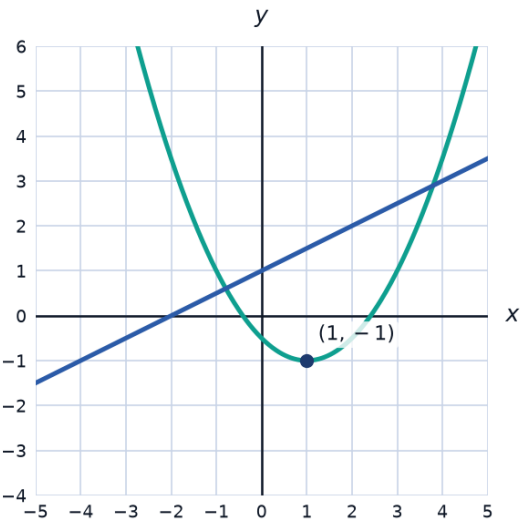
Consider a dataset of observations $\{x_i\}_{i=1}^{n}$ where each x_i is a 2-dimensional vector. The sample covariance matrix for this dataset is given by: $S = (4, -2; -2, 1)$ Principal Component Analysis (PCA) is applied to this dataset. The first principal component corresponds to the direction with the largest variance. Let the direction vector of the first principal component be $\mathbf{v}_1 = (a; b)$. The variance explained by this principal component is given by the corresponding eigenvalue λ_1. If $\mathbf{v}_1$ is a unit vector, what is the value of $a^2 + b^2$?

5 [Foundational] • Target: High-Dimensional Geometry and Projections

A linear transformation is defined by the matrix $M = (2, 1; 3, 4)$. If an input vector is $v = (x; 1)$, and the output vector after the transformation is $w = (5; 7)$, what is the value of x?

6 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Consider a linear layer with a weight matrix W and an input vector $\mathbf{x}$. If $W = (1, 2; 3, 4)$ and $\mathbf{x} = (5; 6)$, compute the output vector $\mathbf{y} = W\mathbf{x}$.



7 [Foundational] • Target: Vector Spaces and Linear Combinations

A vector $\mathbf{v}$ is transformed into a new vector $\mathbf{v}'$ by a linear transformation represented by the matrix $\mathbf{M}$. We are given that $\mathbf{v} = (2\ 3)$ and $\mathbf{v}' = (7\ 11)$. The transformation matrix is of the form $\mathbf{M} = (a, 1\ 4, b)$. Determine the value of a and b.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Vector Norms and Sparsity

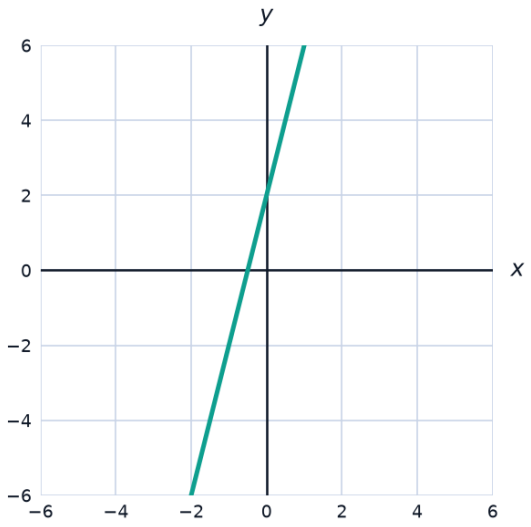
Consider two vectors, $\mathbf{u} = [3; 4]$ and $\mathbf{v} = [1; -2]$. What is the vector representing $2\mathbf{u} - \mathbf{v}$?

9 [Intermediate] • Target: Dot Products, Cosine Similarity and Angular Distance

Consider two vectors, $\mathbf{u} = (3; -2)$ and $\mathbf{v} = (-1; 5)$. What is the resultant vector $\mathbf{w} = \mathbf{u} + \mathbf{v}$?

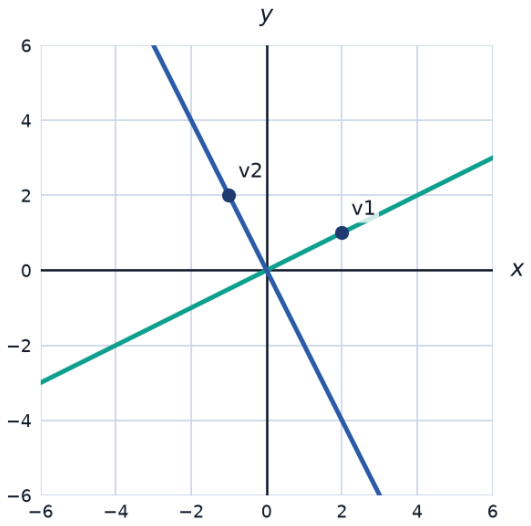
10 [Intermediate] • Target: High-Dimensional Geometry and Projections

A simple linear regression model has been fitted to a dataset. The model equation is given by $\hat{y} = ax + b$. The regression line passes through the points $(2, 10)$ and $(5, 22)$ in the feature-target space. Determine the value of the intercept b.



11 [Intermediate] • Target: Vectors & Embeddings in AI Practice Exercises

A 2D linear transformation T is defined by the matrix $M = (3, 1; 1, 3)$. Consider a triangle with vertices $A = (1, 0)$, $B = (0, 1)$, and $C = (1, 1)$. This triangle is transformed by T to a new triangle with vertices A', B', and C'. The area of the transformed triangle $A'B'C'$ is then passed through a ReLU activation function, $f(x) = \max(0, x)$. What is the final output value?



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12 [Intermediate] • Target: Vector Spaces and Linear Combinations

Consider a 2D dataset represented by the points $A = (1, 2)$, $B = (4, 2)$, and $C = (2, 5)$. If this dataset undergoes a linear transformation represented by the matrix $M = (2, 0; 0, 1)$, what are the coordinates of the transformed points A', B', and C'?

13 [Intermediate] • Target: Vector Norms and Sparsity

A binary classification model uses a logistic sigmoid function $\sigma(z) = \frac{1}{1+e^{-z}}$. The decision boundary is defined by $\sigma(z) = 0.5$, where $z = w_1x_1 + w_0$. For a dataset with two features x_1 and x_2, the model's linear component is $z = 2x_1 - x_2 + b$. If the classification boundary passes through the point $(x_1, x_2) = (3, 1)$, what is the value of the bias b?

14 [Intermediate] • Target: Dot Products, Cosine Similarity and Angular Distance

Consider a simple linear regression model where the predicted value $\hat{y}$ is given by $\hat{y} = b_0 + b_1x$. If we have two data points $(x_1, y_1) = (2, 5)$ and $(x_2, y_2) = (4, 9)$, and the Ordinary Least Squares (OLS) method yields the regression line $\hat{y} = 1 + 2x$, what is the predicted value $\hat{y}$ for $x = 3$?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

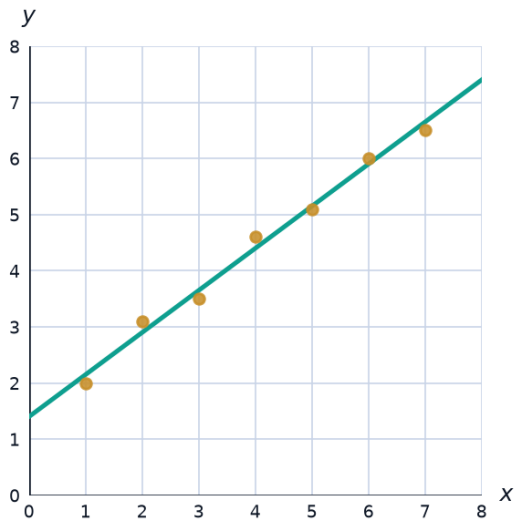
15 [Challenging] • Target: High-Dimensional Geometry and Projections • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Consider a simple logistic regression model where the decision boundary is defined by the linear equation $w_1x_1 + w_2x_2 + b = 0$. If the weights are $w_1 = 2$ and $w_2 = 3$, and the bias is $b = -7$, what is the equation of the decision boundary?

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16 [Challenging] • Target: Vectors & Embeddings in AI Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Consider a simple linear regression model where a dependent variable y is predicted from an independent variable x. The model is given by the equation $y = \beta_0 + \beta_1 x + \varepsilon$, where ε is the error term. Suppose we have estimated the regression line to be $\hat{y} = 3 + 2x$. If we observe an independent variable value of $x = 4$, what is the predicted value $\hat{y}$ for the dependent variable?



17 [Challenging] • Target: Vector Spaces and Linear Combinations • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 20$ and the Total Sum of Squares is $\text{SST} = 100$. What is the coefficient of determination R^2?

18 [Challenging] • Target: Vector Norms and Sparsity • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

A vector $\mathbf{v}$ is represented in a 2D space. If the components of the vector are $v_x = -3$ and $v_y = 4$, what is the Euclidean (L_2) norm of this vector?

19 [Challenging] • Target: Dot Products, Cosine Similarity and Angular Distance • Cross-Lesson: 6.01 Linear Regression and OLS

Consider a 2D data point $\mathbf{v} = [3;\ -4]$. What is the squared Euclidean norm (or squared magnitude) of this vector?

20 [Challenging] • Target: High-Dimensional Geometry and Projections • Cross-Lesson: 7.01 Logistic Regression and Classification

A 2D loss surface is defined as $L(w_1, w_2) = 2w_1^2 + 3w_2^2$. Find the gradient vector $ablaL$ at the point $(w_1, w_2) = (2,\ -1)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Compute the dot product of the embedding vector \(\mathbf{u} = [2, 3]^T \) with itself.

a. \(13 \)

b. \(15 \)

c. \(11 \)

d. \(10 \)

22 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Compute the dot product of the embedding vector \(\mathbf{u} = [3, 4]^T \) with itself.

a. \(25 \)

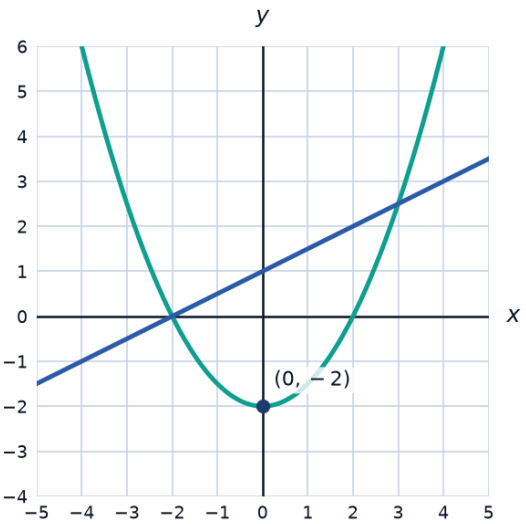
b. \(29 \)

c. \(23 \)

d. \(14 \)

23 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Compute the dot product of the embedding vector \(\mathbf{u} = [4, 5]^T \) with itself.



a. \(41 \)

b. \(47 \)

c. \(39 \)

d. \(18 \)

24 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Compute the dot product of the embedding vector \(\mathbf{u} = [5, 6]^T \) with itself.

a. \(61 \)

b. \(69 \)

c. \(59 \)

d. \(22 \)

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25 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Compute the dot product of the embedding vector $\mathbf{u} = [6, 7]^T$ with itself.

a. 85

b. 95

c. 83

d. 26

26 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Compute the dot product of the embedding vector $\mathbf{u} = [7, 8]^T$ with itself.

a. 111

b. 125

c. 113

d. 30

27 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Compute the Manhattan (L_1) norm and Euclidean (L_2) norm of the word vector $\mathbf{w} = [2, -3]^T$.

a. $\|\mathbf{w}\|_1 = 5, \; \|\mathbf{w}\|_2 = \sqrt{13}$

b. $\|\mathbf{w}\|_1 = -1, \; \|\mathbf{w}\|_2 = 13$

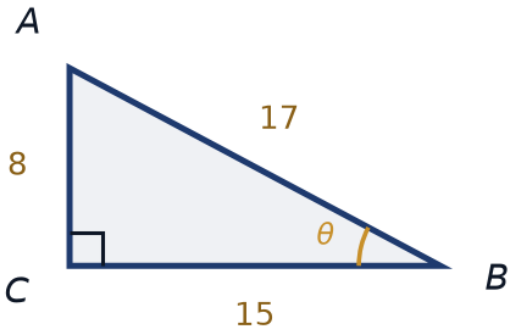
c. $\|\mathbf{w}\|_1 = 5, \; \|\mathbf{w}\|_2 = 5$

d. $\|\mathbf{w}\|_1 = \sqrt{13}, \; \|\mathbf{w}\|_2 = 5$

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28 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Compute the Manhattan (\(L_1 \)) norm and Euclidean (\(L_2 \)) norm of the word vector \(\mathbf{w} = [4, -6]^T \).



a. \(\|\mathbf{w}\|_1 = -2, \; \|\mathbf{w}\|_2 = 52 \)

b. \(\|\mathbf{w}\|_1 = 10, \; \|\mathbf{w}\|_2 = \sqrt{52} \)

c. \(\|\mathbf{w}\|_1 = 10, \; \|\mathbf{w}\|_2 = 10 \)

d. \(\|\mathbf{w}\|_1 = \sqrt{52}, \; \|\mathbf{w}\|_2 = 10 \)

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Vector Spaces and Linear Combinations

Given word vectors $\mathbf{u} = [2, -1, 3]^T$ and $\mathbf{v} = [1, 4, -2]^T$, compute the linear combination $3\mathbf{u} - 2\mathbf{v}$.

a. $[4, -11, 13]^T$

b. $[4, 5, 5]^T$

c. $[8, -11, 5]^T$

d. $[7, 5, 13]^T$

30 [Intermediate] • Target: Vector Spaces and Linear Combinations

Two embedding vectors $\mathbf{a} = [k, 2, -1]^T$ and $\mathbf{b} = [3, -4, 2]^T$ are orthogonal in semantic space. Find the value of k.

a. $k = \frac{10}{3}$

b. $k = 2$

c. $k = -\frac{10}{3}$

d. $k = 5$

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31 [Intermediate] • Target: Vector Spaces and Linear Combinations

Determine whether the three feature vectors $\mathbf{v}_1 = [1, 0]^T$, $\mathbf{v}_2 = [0, 1]^T$, and $\mathbf{v}_3 = [3, -2]^T$ are linearly independent in $\mathbf{R}^2$.

a. Linearly dependent (any set of > 2 vectors in $\mathbf{R}^2$ is dependent)

b. Linearly independent

c. Orthogonal set

d. Cannot be determined without a matrix inversion

32 [Intermediate] • Target: Vector Norms and Sparsity

Calculate the Manhattan (L_1) norm and Euclidean (L_2) norm of the embedding error vector $\mathbf{e} = [3, -4, 12]^T$.

a. $||\mathbf{e}||_1 = 19, \quad ||\mathbf{e}||_2 = 13$

b. $||\mathbf{e}||_1 = 13, \quad ||\mathbf{e}||_2 = 19$

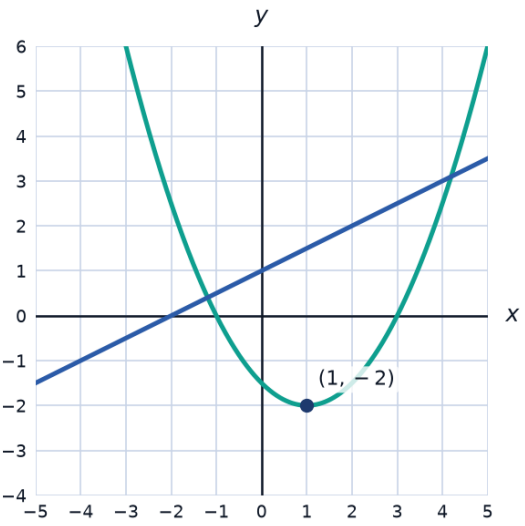
c. $||\mathbf{e}||_1 = 11, \quad ||\mathbf{e}||_2 = 13$

d. $||\mathbf{e}||_1 = 19, \quad ||\mathbf{e}||_2 = 169$

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33 [Intermediate] • Target: Vector Norms and Sparsity

Find the Chebyshev (L_∞) norm of the parameter vector $\mathbf{w} = [-7, 2, 5, -9, 4]^T$.



a. 9

b. 7

c. 27

d. 5

34 [Intermediate] • Target: Vector Norms and Sparsity

In Lasso regression, why does adding an L_1 penalty to the loss function induce exact feature sparsity (weights driven to 0)?

a. The diamond-shaped L_1 contour has sharp corners on the coordinate axes

b. The L_1 norm is continuously differentiable everywhere

c. The L_1 penalty penalizes large weights much more heavily than L_2

d. The L_1 norm computes the determinant of the covariance matrix

35 [Intermediate] • Target: Dot Products, Cosine Similarity and Angular Distance

Two query embeddings are $\mathbf{u} = [1, 1, 0]^T$ and $\mathbf{v} = [0, 2, 2]^T$. Compute the cosine similarity $\cos\theta$ between them.

a. 0.500

b. 0.707

c. 0.250

d. 1.000

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36 [Intermediate] • Target: Dot Products, Cosine Similarity and Angular Distance

If two unit-normalized embeddings have cosine similarity $\cos\theta = -1.0$, what is their semantic relationship?

a. Diametrically opposite in direction (180° apart)

b. Identical in meaning

c. Orthogonal / unrelated

d. Collinear in the exact same direction

37 [Intermediate] • Target: Dot Products, Cosine Similarity and Angular Distance

Compute the cosine distance $D_{\cos}(\mathbf{u}, \mathbf{v}) = 1 - \cos\theta$ for two orthogonal feature vectors.

a. 1.0

b. 0.0

c. 2.0

d. -1.0

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: High-Dimensional Geometry and Projections • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

Find the scalar projection of vector $\mathbf{a} = [4, 2]^T$ onto the direction of unit vector $\mathbf{u} = [0.6, 0.8]^T$.

a. 4.0

b. 5.0

c. 2.4

d. 3.2

39 [Challenging] • Target: High-Dimensional Geometry and Projections • Cross-Lesson: 6.01 Linear Regression and OLS

In high-dimensional embedding spaces ($D = 1024$), why are two randomly selected vectors almost certainly nearly orthogonal?

a. The surface area of a high-dimensional hypersphere concentrates near the equator relative to any pole

b. All high-dimensional vectors have zero norm

c. The dot product of random Gaussian variables always equals 1

d. The cosine function approaches infinity as dimension increases

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40 [Challenging] • Target: High-Dimensional Geometry and Projections • Cross-Lesson: 7.01 Logistic Regression and Classification

Normalize the feature vector $\mathbf{x} = [3, 0, 4]^T$ to have unit Euclidean norm.

a. $[0.6, 0, 0.8]^T$

b. $[0.75, 0, 1]^T$

c. $[1, 0, 1]^T$

d. $[0.3, 0, 0.4]^T$

41 [Challenging] • Target: Vectors & Embeddings in AI Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Two document embeddings $\mathbf{a} = [6, 12]^T$ and $\mathbf{b} = [12, 6]^T$ are compared. If $||\mathbf{a}|| = ||\mathbf{b}|| = \sqrt{180}$, calculate their exact cosine similarity.

a. 1.000

b. 0.640

c. 0.800

d. 0.500

42 [Challenging] • Target: Vector Spaces and Linear Combinations • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Two document embeddings $\mathbf{a} = [2, 4]^T$ and $\mathbf{b} = [4, 2]^T$ are compared. If $||\mathbf{a}|| = ||\mathbf{b}|| = \sqrt{20}$, calculate their exact cosine similarity.

a. 1.000

b. 0.640

c. 0.800

d. 0.500

43 [Challenging] • Target: Vector Norms and Sparsity • Cross-Lesson: 3.01 Gradient Descent and Optimization

Two document embeddings $\mathbf{a} = [3, 6]^T$ and $\mathbf{b} = [6, 3]^T$ are compared. If $||\mathbf{a}|| = ||\mathbf{b}|| = \sqrt{45}$, calculate their exact cosine similarity.

a. 1.000

b. 0.640

c. 0.800

d. 0.500

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44 [Challenging] • Target: Dot Products, Cosine Similarity and Angular Distance • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Two document embeddings $\mathbf{a} = [4, 8]^T$ and $\mathbf{b} = [8, 4]^T$ are compared. If $||\mathbf{a}|| = ||\mathbf{b}|| = \sqrt{80}$, calculate their exact cosine similarity.

a. 1.000

b. 0.640

c. 0.800

d. 0.500

45 [Challenging] • Target: High-Dimensional Geometry and Projections • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

Two document embeddings $\mathbf{a} = [1, 2]^T$ and $\mathbf{b} = [2, 1]^T$ are compared. If $||\mathbf{a}|| = ||\mathbf{b}|| = \sqrt{5}$, calculate their exact cosine similarity.

a. 1.000

b. 0.640

c. 0.800

d. 0.500

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